

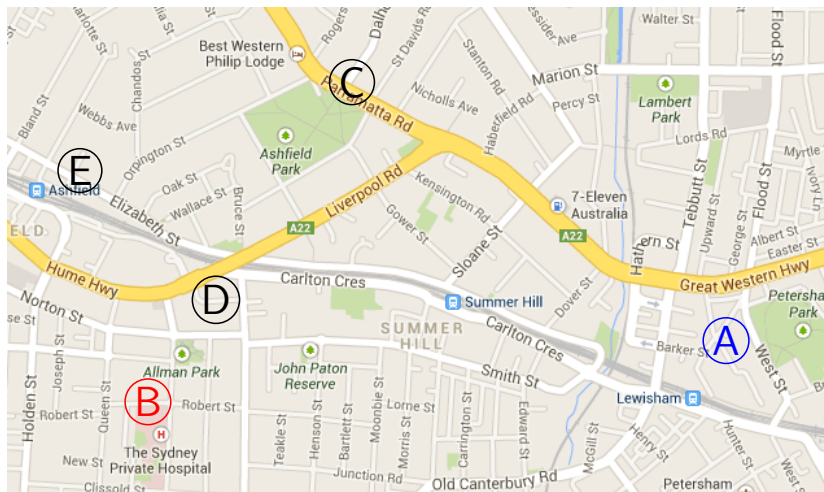
Lesson 1.2: Decision Elements

Introduction to Decision-making

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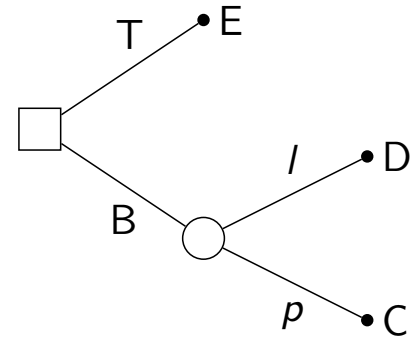
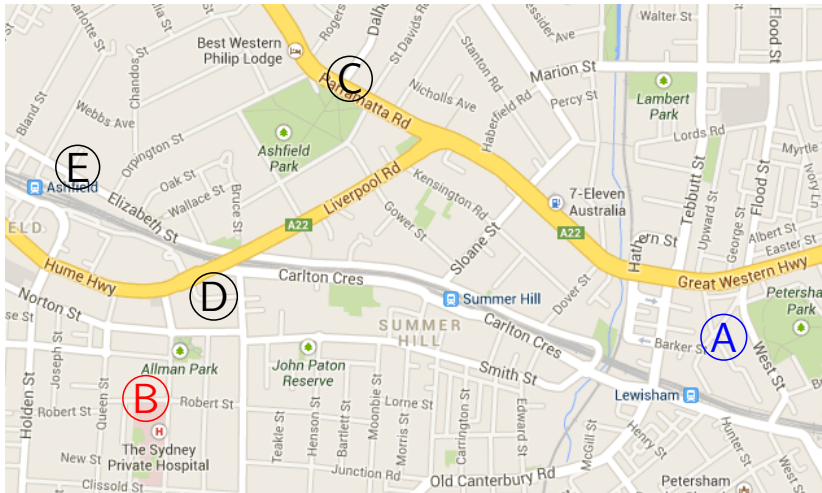
Decision elements



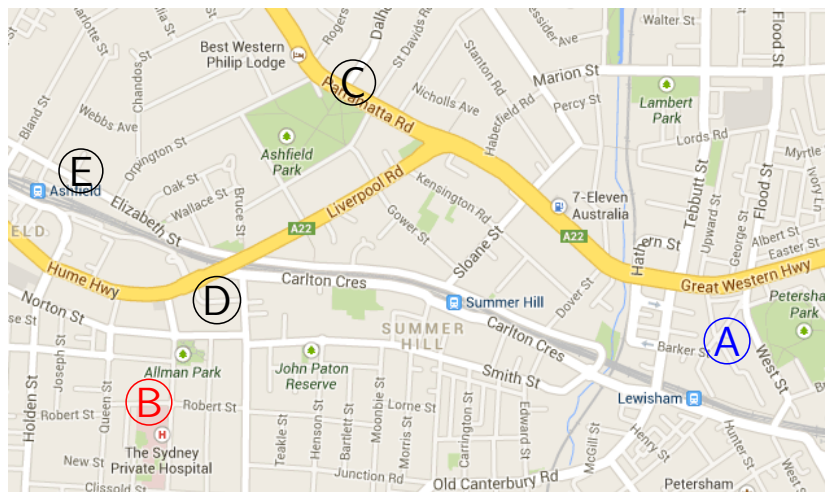
Example (Trip planning)

You have to get from Petersham Park (A) to the Hospital (B) by either **train** or **bus**. The train goes to Ashfield Station (E). You don't know the bus route: either via **Parramatta rd** (taking you to C) or **Liverpool rd** (taking you to D).

Tree model/representation



Decision elements (cont'd)



Basic elements common to decisions:

actions (alternatives, options) ($\mathcal{A}$): T, B

states (cases) ($\mathcal{S}$): I, p

outcomes (consequences) (Ω): arrive at C, D, or E

Oil prospecting

Example (To drill or not to drill?)

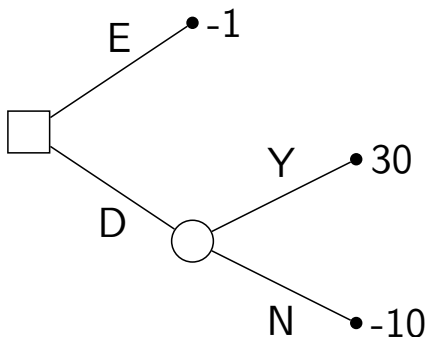
Your petroleum company has paid for a lease an oil site. Unfortunately, investigations have been inconclusive as to whether or not there's **oil present** at the site. The cost of the lease was much less than the cost of drilling for oil, which is very expensive. However, if you **drill** and find oil, you stand to make a huge profit. Alternatively, you may let the lease **expire** without drilling.

actions Drill (D), Expire (E)

states oil present (o), no oil ($\bar{o}$)

outcomes huge profit, big loss, small loss

Abstraction



	Trip planning	Oil exploration
C	T	E
U	B	D
p	Liverpool rd	oil present
c	location E	small loss
b	location D	huge profit
w	location C	big loss